

Program 1

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ZZJOB 5                                CSC 001    GERALD, C. F.
ZZFORX5
*LIST PRINTER
C SOLUTION TO F(X) = 0 BY INTERVAL HALVING METHOD.
C CHANGE FUNCTION DEFINITION CARD TO SOLVE NEW EQUATION.
  F(X) = EXPF(-X) - SIN(3.1415926*X/2.)
C READ IN X1, X2 VALUES, TOL.
  READ 100, X1, X2, TOL
100 FORMAT (2F10.0, E10.0)
C COMPUTE F(X1), F(X2) AND TEST IF OPPOSITE SIGNS OR ZERO.
  FX1 = F(X1)
  FX2 = F(X2)
  IF (FX1*FX2) 1, 2, 3
3 TYPE 200
200 FORMAT (80H FUNCTION VALUES WITH INITIAL X VALUES ARE NOT OF OPPOS
  ITE SIGNS. CHANGE INPUTS. )
  CALL EXIT
C WHEN PRODUCT OF FX1*FX2 IS ZERO, X1 OR X2 IS A ROOT.
2 IF (FX1) 4, 5, 4
5 PRINT 201, X1
201 FORMAT (22H THE INPUT VALUE, X = ,F6.3, 27H IS A ROOT OF THE EQUAT
  ION. )
4 IF (FX2) 6, 7, 6
7 PRINT 201, X2
  CALL EXIT
6 PRINT 202
202 FORMAT (55H ERROR. FX1 AND FX2 NOT ZERO THOUGH PRODUCT TESTS ZERO.
  1 )
  CALL EXIT
C BEGIN INTERVAL HALVING PROCESS. LIMIT TO 25 TIMES.
1 DO 30 I = 1, 25
  X = (X1 + X2)/2.
  FX = F(X)
  PRINT 206, X, FX
206 FORMAT (8H AT X = , F10.7, 8H F(X) = ,F10.6)
  IF (ABS(FX) - TOL) 8, 8, 9
C TEST WHETHER F(X) AND F(X1) ARE OF OPPOSITE SIGNS.
C RESET VALUES ACCORDINGLY.
9 IF (FX*FX1) 10, 11, 12
10 X2 = X
  GO TO 30
12 X1 = X
  FX1 = FX
30 CONTINUE
  PRINT 204
204 FORMAT (33H NON-CONVERGENT IN 25 ITERATIONS. )
  CALL EXIT
11 PRINT 205
205 FORMAT (7H ERROR. )
  CALL EXIT
8 PRINT 203, X, 1
203 FORMAT (16H THE VALUE, X = ,F10.7, 12H IS A ROOT. , 13, 26H ITERAT
  IONS WERE REQUIRED. )
  CALL EXIT
END
0.      1.      1.E-5
ZZZZ
AT X = .5000000 F(X) = -.100576
AT X = .2500000 F(X) = .396117
AT X = .3750000 F(X) = .131719
AT X = .4375000 F(X) = .011255
AT X = .4687500 F(X) = -.045774
AT X = .4531250 F(X) = -.017534
AT X = .4453125 F(X) = -.003207
AT X = .4414062 F(X) = .004006
AT X = .4433593 F(X) = .000395
AT X = .4443359 F(X) = -.001407
AT X = .4438476 F(X) = -.000506
AT X = .4436035 F(X) = -.000055
AT X = .4434814 F(X) = .000170
AT X = .4435424 F(X) = .000057
AT X = .4435729 F(X) = .000001
THE VALUE, X = .4435729 IS A ROOT. 15 ITERATIONS WERE REQUIRED.

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